

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Three genes(1) Any two (x1) from: There are three genes present (1) (proteins in) glycoprotein (1) capsid (is made from protein) (1) reverse transcriptase (is an enzyme made of protein) (1)			3	3		
		(ii)		Any two (x1) from: - It does not have any ribosomes {to translate mRNA / for protein synthesis} (1) - Does not have mitochondria to produce ATP (1) - Host cell needed to supply raw materials (e.g. free nucleotides/ amino acids) / OWTTE (1)		2		2		
	(b)	(i)		2 marks for 45 216 (if use 3.14) 45 239 (if use π on calculator) Accept any number of DPs if using π on calculator 45238.9342 (rounded correctly) If incorrect award 1 mark $4 \times \{3.14 \text{ or } \pi\} \times 60^2$ 180864 (using 120nm as radius + 3.14) 180955.7368 (using 120nm as radius + π)		2		2	2	
		(ii)		A - endocytosis and B - exocytosis (1) A results in {membrane / surface area} being {added / increased} and B results in {membrane / surface area} being {lost / decreased} (1)		1	1	2		
				Question 5 total	0	5	4	9	2	